ChatGPT

At-Cost Pricing for Maximum Distribution with a Separate Bounty for Problem Solving

Executive summary

Verdict. The proposal is economically coherent, but it is not a replacement for the whole market system in its present form. Most of its components have strong prior art. The distinctive contribution is the *combination*: recursively passing through audited costs, separating use-caused costs from shared capacity costs, treating interfaces as common infrastructure, removing innovation rents from the access price, and paying a separately governed reward for solving the underlying problem. I found no primary source that puts all of those pieces together in that form. That is evidence that the architecture is partly novel, not proof of priority.

The welfare case for the core separation is unusually strong for information goods. Arrow made almost exactly the central observation in 1962: once information exists, efficient use calls for making it available at no more than the cost of transmitting it, but doing so destroys the private incentive to create it. He therefore treated *rewarding invention* and *charging users for the information* as conceptually separable problems. [1] Hotelling had earlier made the analogous case for marginal-cost pricing of infrastructure with increasing returns, while Coase showed why this does not by itself solve fixed-cost recovery, investment choice, or demand revelation. [2]

The proposal is therefore **well established in principle** on four points. Monopoly prices can create deadweight loss by excluding mutually beneficial transactions; that exclusion is especially stark for non-rival information once it has been created. [3] Recursive cost propagation has a rigorous basis in Leontief input-output systems, subject to productivity conditions of the Hawkins–Simon type. [4] Innovation can be rewarded separately from the access price through prizes, buyouts, advance commitments, impact payments, procurement, grants, open-source funding and related mechanisms. [5] And organizations operating broadly on a “service at cost” basis already exist, especially cooperatives and regulated infrastructure, although they generally retain depreciation, reserves and some return on capital rather than interpreting “no profit” literally. [6]

What is **partly new** is turning these strands into an interoperable pricing protocol across organizations: each layer reports direct resource cost plus purchased inputs; common interfaces are financed outside transaction-level prices; residual capacity costs are allocated separately; and innovation rewards are shown beside, rather than hidden inside, prices. The Billera–Heath–Raanan application of Aumann–Shapley pricing to Cornell's internal telephone system is surprisingly close on the shared-cost side: they explicitly sought rates that were fair, covered all costs, and handled large fixed and nonlinear costs shared among many small users. [7] Agricultural cooperatives are close on the accounting side: because exact costs cannot be known in advance, they can charge provisional prices and later return net margins according to patronage, effectively making “at cost” an ex-post settlement rather than requiring perfect foresight. [8]

What appears **genuinely distinctive**, within the literature checked here, is the proposed package of a recursively composable at-cost layer **plus** near-zero interface tolls **plus** an explicitly demand- or impact-related bounty **plus** a traceable distributive ledger. Arrow anticipated the central price/reward separation. Leontief anticipated recursive accounting. Cooperatives anticipated at-cost settlement. Prize theory anticipated separate rewards. Ostrom anticipated nested governance. But I found no close predecessor that treats the four as one general economic protocol. That novelty claim should remain provisional until a broader patent, mechanism-design and accounting-standards search is done.

The strongest objection is also clear. **Profit is not only an innovation bounty.** It can simultaneously reward risk-bearing, finance experimentation, provide a residual claim that creates incentives to economize, signal scarcity and unexpected demand, fund renewal, and determine which projects survive. A system that removes profit from price has to reproduce those functions somewhere else. Hayek's objection is particularly important: computing an internally consistent cost vector is not the same thing as discovering what should be produced, in what quality, where, when, or by whom. The relevant knowledge is dispersed and often created through the competitive process itself. [9] Cost-based regulation adds another problem: if a producer can recover whatever cost it reports, it can have too little incentive to reduce costs and, under some regulatory systems, an incentive to use too much capital. Averch–Johnson and Laffont–Tirole formalize different versions of this problem. [10]

An explicit bounty **helps but does not solve this by itself**. Quantity demanded at an at-cost price does not reveal total value. It only shows that users value another unit at least as much as that low price. A free or nearly free product can have enormous use and modest social value, or little measured use and enormous option or public value. A viable bounty therefore needs additional value-revelation devices: competitive bids, scarce commitment budgets, outcome measurement, counterfactual impact analysis, independent evaluation, or some combination. Kremer's patent auctions, advance market commitments and the Health Impact Fund all exist precisely because simple observed sales do not supply the missing counterfactual. [11]

The pneumococcal-vaccine Advance Market Commitment is an important warning against elegant theory becoming an automatic success story. The final Gavi evaluation found that the AMC helped scale supply and accelerated uptake, but found no evidence that it accelerated R&D among products already in the pipeline; it also experienced early shortages, over-optimistic demand forecasts and weak price competition. [12] The lesson is favorable to the architecture but hostile to simple formulas: separate rewards can work, but the counterfactual, market design and implementation details dominate outcomes.

The best use case is therefore not immediate economy-wide replacement of ordinary prices. It is a **voluntary layer inside a market economy**, beginning where marginal reproduction cost is low, costs can be audited, quality can be measured, suppliers remain contestable, and an institution can credibly fund the bounty and commons. Open-source public-sector software is a particularly good first pilot.

Research findings across the ten questions

Question one — The cost of restricting supply

Evidence. In the standard partial-equilibrium monopoly model, a seller with market power chooses a price above marginal cost and output below the competitive quantity. The resulting lost trades form the conventional Harberger deadweight-loss triangle. Harberger's 1954 exercise famously produced a very small aggregate estimate for US manufacturing: under his assumptions, eliminating the measured resource

misallocation would have generated welfare gains on the order of one-tenth of one percent of national income. He was explicit that this was an allocative calculation that abstracted from distribution. [13]

That estimate did not settle the issue. Later work changed the unit of observation, demand assumptions and definition of social cost. Cowling and Mueller argued that industry aggregation and treatment of monopoly rents could make conventional estimates too low. Posner went further conceptually: resources spent acquiring, defending or obtaining monopoly rents can themselves be social costs, rather than mere transfers. [14] More recently, De Loecker, Eeckhout and Unger estimated that aggregate US markups rose from about 21 percent above marginal cost around 1980 to 61 percent in the later part of their sample, driven mainly by the upper tail rather than by a rising median markup. That is evidence about market power, not a direct estimate of deadweight loss; translating markups into welfare requires assumptions about demand, entry, fixed costs and productivity. [15]

So there is no credible single economy-wide “percentage lost to monopoly” that can be carried from Harberger to the present. Harberger's small number was conditional on a deliberately narrow static exercise. Broader estimates count different phenomena. Modern markup estimates establish that the empirical premises have changed, but not the welfare number by themselves. [16]

Arrow's information argument is much closer to the present proposal. Information is non-rival: another person using a discovered formula, software routine or fact normally does not prevent the first person from using it. Once the information has been created, Arrow argued that optimal utilization requires charging no more than its transmission cost; doing so, however, undermines the incentive to incur the fixed cost of discovery. He explicitly described an ideal arrangement in which the inventor's reward is separated from what users pay for the information. [17] This does **not** mean the total welfare stakes of every non-rival product are necessarily larger than those of every physical monopoly. It means that, conditional on creation, the marginal social cost of additional use may be extremely low, making exclusion particularly hard to justify on allocative grounds.

Price discrimination complicates the story. Third-degree discrimination can open markets that would otherwise not be served, but it can also raise prices to groups with relatively inelastic demand. Schmalensee and Varian showed that its welfare effect is generally ambiguous; an increase in total output is an important condition for it to improve conventional surplus measures. [18] Perfect first-degree discrimination is more revealing: in the textbook model it can eliminate the conventional output deadweight loss while transferring almost all surplus to the seller. That separates **efficiency** from **distribution**. A tiered price can therefore improve access without making the distribution of gains neutral.

Inference. The proposal's claim that restricting low-marginal-cost goods can destroy substantial value is established theory. Its stronger ethical premise—that maximizing distribution should replace profit maximization—is a value judgment, not an implication of welfare economics. A useful design should state that openly.

Question two — Economy-wide at-cost price systems

Evidence. Leontief gives the cleanest mathematical foundation for recursively composed prices. Let A be the matrix of physical intermediate-input requirements and let v be direct primary costs per unit. An at-cost price vector can be written

$$p = A^\top p + v,$$

so that

$$p = (I - A^\top)^{-1} v.$$

Each price therefore contains the direct primary cost of its own production plus the recursively embodied cost of every purchased input. That is almost exactly the accounting operation envisaged in the proposal. [19]

Existence is not automatic. For a non-negative input matrix, the Hawkins–Simon productivity conditions give conditions under which the economy can produce positive net output and the Leontief inverse is economically meaningful. In spectral language, the familiar sufficient formulation is $\rho(A) < 1$: the embodied input requirements shrink rather than explode as recursion proceeds. [20] Under a fixed linear technology, that condition also makes ordinary fixed-point iteration converge. But this should not be overstated. Static existence of a price solution does not establish stability when technologies, quantities, capacities, expectations or coefficients themselves react to prices.

Sraffa is important precisely because he shows what the word **cost** fails to settle. Prices of production depend not only on intermediate commodities but on how the economic surplus is divided between wages and profits. A price vector is therefore not generally unique merely because the production technique is known; a distributive variable and a numeraire must also be fixed. [21] Setting the conventional profit rate to zero can create an at-cost special case, but it does not make capital free. Replacement, depreciation, risk-bearing and any chosen opportunity cost of scarce capital still have to be assigned somewhere.

This matters for the proposal's claim of neutrality. An “at-cost” economy has to answer at least four distributive questions before its prices are determined: what labor costs count; how capital formation and depreciation count; which taxes or common charges count; and whether external damage counts. Those are accounting and ethical choices, not outputs of the recursion.

National supply-use and input-output tables provide a practical benchmark. Eurostat's framework records inter-industry/product flows, value added, imports and the relationships between basic and purchaser prices, and supports computation of Leontief inverses. [22] Such tables could supply *reference coefficients* for suspiciously high or low reported costs and estimate embodied upstream cost. They cannot function as real-time invoices. They are aggregated, retrospective and based on statistical conventions; they do not know a particular supplier's technology, quality, spare capacity or next investment.

Inference. Recursive at-cost prices can exist and be unique under explicit productivity and accounting assumptions. An economy-wide system is mathematically possible in the same limited sense that an input-output system is solvable. What does **not** follow is that those prices contain all information needed for decentralized allocation. They are accounting prices. Scarcity and discovery remain separate problems.

Question three — Separating the bounty from the price

Several major lines of research are very close to the proposal.

Kremer's 1998 patent-buyout mechanism proposed using an auction to estimate the private value of a patent, multiplying the estimate to reflect the gap between private and social value, buying most selected patents into the public domain, and occasionally reselling one to create incentives for bidders to state genuine valuations. The attraction is exactly the one at issue here: preserve the innovation reward without preserving the monopoly-use distortion. Kremer also identified collusion and manipulation as central risks. [23]

Shavell and van Ypersele formally compared intellectual-property rights with government rewards. In their model, rewards can outperform monopoly rights because the innovation can enter the public domain, although the government then acquires the hard informational problem of determining the correct reward. [24] Stiglitz has likewise argued for prize mechanisms in medicines so that innovators can be rewarded while products are sold competitively rather than through monopoly pricing. [25]

The Health Impact Fund is closer still. Hollis and Pogge propose an optional arrangement under which a pharmaceutical innovator receives payments based on measured global health impact while accepting a low product price. Their design envisages annual assessment of incremental health impact and, in the version documented in the 2008 proposal, a price no higher than assessed long-run marginal production and distribution cost. [26] Its central difficulty is also visible in the proposal itself: measuring incremental health impact across countries and diseases requires extensive data, counterfactuals and protection against reward-seeking behavior that does not actually improve health. [27]

Advance market commitments approach the problem ex ante. Rather than rewarding whatever happens to be produced, a sponsor commits in advance to a market or subsidy conditional on products meeting an agreed specification. The pneumococcal AMC committed donor funds to stimulate supply of vaccines suitable for lower-income countries while constraining the long-run price. [28] Its final evaluation is sobering and valuable: Gavi reported strong effects on scaling supply and accelerated uptake, but no evidence of acceleration of pipeline R&D, early supply shortages, optimistic demand forecasts and relatively weak price competition. [12]

Open-source production demonstrates a different family of rewards. Lerner and Tirole showed that contributors can receive career, reputation and complementary commercial benefits even when source code itself is freely available. [29] Benkler's peer-production analysis makes the broader point that some information production can be coordinated outside ordinary firm and price hierarchies. [30] Quadratic funding offers a more formal way for a common fund to amplify many small expressions of support for public goods, though its attractive properties depend on assumptions about independent contributors and require defenses against collusion and false identities. [31]

The central measurement result is negative: ordinary demand at an at-cost price cannot by itself size the bounty. Suppose software costs effectively zero for one more user. Ten million downloads establish that ten million users prefer it to a price close to zero. They do not reveal whether aggregate willingness to pay is $1 million, $100 million or $10 billion. Nor do they establish how much of the observed benefit would have existed without the innovation.

A credible bounty therefore needs a second signal. The literature supplies at least four candidates: auctioned valuations, as in Kremer; binding advance purchase commitments; independently measured outcomes, as in the HIF; and decentralized contributions with matching, as in quadratic funding. [32]

Inference. Separating reward from price is not novel and is theoretically well founded. What remains unresolved is the hardest part: obtaining a reward signal that is both informative and difficult to game.

Question four — Knowledge, incentives and discovery

Mises's 1920 calculation argument was directed at an economy without markets in productive assets. Without exchange prices for capital goods, he argued, planners lack the monetary comparisons needed to choose among technically possible production plans. [33] A voluntary at-cost network embedded in a market does not face the objection in its strongest form: it can buy labor, energy, machines and other scarce inputs at outside market prices. An economy that attempted to replace *all* such markets with recursively reported historical costs would face the objection much more directly.

Hayek's 1945 argument goes deeper. The relevant facts are not all “given” to one mind. Much useful knowledge concerns particular circumstances of time and place and remains dispersed among people. Prices help transmit some consequences of that dispersed knowledge without requiring anyone to assemble it centrally. [34] Solving a larger Leontief matrix with modern computers does not answer this objection. Computation can solve a stated optimization problem faster; it cannot automatically generate unknown preferences, entrepreneurial ideas, local quality judgments, hidden production possibilities or future innovations.

Lange's market-socialist reply conceded the informational role of prices but proposed that a central board could set accounting prices and adjust them when shortages or surpluses appeared while instructing managers to follow marginal-cost and least-cost rules. [35] That reply exposes a tension in the present proposal. When capacity is scarce, a price that is required always to equal reported accounting cost no longer communicates scarcity. The system must then ration by queue, priority, auction, waiting time, central decision, or some other mechanism. If it raises the price until demand matches capacity, it has reintroduced a scarcity price.

Cost-based incentives create a separate problem. Averch and Johnson showed that a rate-of-return regulatory regime can bias a regulated firm toward excessive capital intensity. [36] Laffont and Tirole later treated regulation explicitly as a principal-agent problem in which the regulator observes costs imperfectly while managerial effort and intrinsic efficiency are hidden. Their optimal contracts trade cost reimbursement against high-powered incentives; pure cost pass-through performs poorly when the supplier can influence the reported cost. [37]

An outcome bounty improves this because it can make revenue depend on value rather than expenditure. But a bounty tied only to adoption can still reward marketing, lock-in, trivial high-frequency uses or manipulation of the measurement boundary. A bounty tied only to estimated social impact can create disputes over counterfactuals and put enormous power in the evaluator.

Inference. An explicit bounty answers only part of the Hayek/incentive objection. A complete design also needs:

1. decentralized freedom to propose solutions and enter the system;
2. genuine competition for the bounty;
3. hard budget constraints and credible exit;
4. a mechanism for scarce capacity that is not disguised as cost accounting;

5. rewards for cost reduction, not reimbursement of every cost incurred;
6. risk capital or milestone finance before the solution exists;
7. independent quality and impact measurement;
8. experimentation, including permission to fail.

Without these, an “at-cost” system risks becoming a cost-plus bureaucracy. With them, it looks less like central planning and more like a market in which *innovation rewards are purchased collectively while the resulting good is distributed at cost*.

Question five — Fixed costs and the commons

Hotelling's 1938 marginal-cost argument provides the cleanest support for taking fixed costs out of marginal user prices. Where declining average costs make marginal-cost revenue insufficient to cover total cost, he favored marginal-cost pricing combined with financing of the deficit through less distortionary general revenue. [38]

Coase's 1946 critique is essential. He argued that one cannot discuss the price of a service separately from the choice of how much infrastructure to build. If users are not confronted with the costs attributable to having the service available, the institution may lose valuable information about whether a project is worth constructing in the first place. [39] Centralizing shared costs therefore removes a transaction friction but also removes one demand-revelation mechanism.

Oi's two-part tariff gives the standard compromise: charge an access fee plus a per-unit price. The per-unit charge can approach marginal cost while the entry charge helps recover fixed cost. [40] But the entry fee itself can exclude light users. It is therefore efficient in a different sense from “maximum distribution.”

Ramsey pricing addresses another version of the same problem. When a system *must* recover a fixed revenue requirement from commodity prices, uniform marginal-cost pricing is infeasible. The second-best solution places relatively larger markups where demand is less responsive, subject to the model's assumptions. [41] Ramsey–Boiteux pricing therefore says something uncomfortable but useful: central funding of common costs is attractive only if the way the common fund itself is raised is less costly or more acceptable than recovering those costs from users.

Buchanan's theory of clubs adds congestion. A shared facility can be non-rival up to a point and congestible beyond it; membership size should therefore balance wider sharing of fixed costs against declining service quality. [42] That is a better model for compute platforms, networks, laboratories and support teams than the pure public-good model.

Ostrom's institutional evidence points away from both atomized private charging and one huge central fund. Durable commons often combine clearly defined responsibilities, monitoring, graduated responses to violations, local participation and, for large systems, **nested enterprises**: governance at multiple levels. [43] Benkler reaches a compatible conclusion for information production: common resources can support substantial decentralized production where tasks are sufficiently modular and contributors have heterogeneous motivations and information. [30]

Inference. “Costs no individual user causes” should not automatically mean “send them to one central payer.” Many such costs are jointly caused by maintaining readiness for a class of users. A robust hierarchy is:

- marginal resource costs to the user who causes them;
- reservation or capacity costs to the group whose peak requirements cause capacity;
- truly common standards, identity, interoperability and basic governance to a wider common fund;
- renewal investment to a protected capital account;
- innovation to the separate bounty mechanism.

That is close to the seed tariff, but it is more precise about causation.

Question six — Externalities and sustainability

Pigou's basic distinction is between private and social net product: when an action imposes benefits or costs that do not enter the decision-maker's accounts, market prices need not produce the social optimum. Taxes, subsidies and other interventions can in principle bring private incentives closer to social costs and benefits. [44]

Leontief showed that pollution can be incorporated into input-output analysis by attaching pollutant generation and abatement to the ordinary production structure. [45] Modern environmentally extended multi-regional input-output databases such as EXIOBASE generalize this idea across countries and supply chains, allowing consumption to be related to embodied emissions, materials, land and other satellite accounts. [46]

So the proposed recursion can technically propagate environmental burdens. Instead of a scalar cost vector, maintain at least two ledgers:

$$\text{financial cost}, \qquad \text{physical externality vector}.$$

The latter might contain kilograms of greenhouse gases, water consumption, particulate emissions and land use. A separate valuation vector converts those physical accounts into money *if* monetary aggregation is wanted. This separation matters because physical tracing is largely an accounting question, whereas assigning €100 rather than €50 to a unit of harm is a social valuation question.

The True Price methodology explicitly adds monetized environmental and social external costs to ordinary market prices to calculate a “true price gap.” Its current methodology illustrates both the possibility and the problem: monetary factors require assumptions about geography, affected populations, damage pathways, valuation methods and normative choices. [47] A “true” price is therefore not a uniquely observable natural constant.

Inference. Yes, a recursive at-cost architecture can carry environmental costs through supply chains. It may actually be unusually well suited to doing so because the chain is explicit. But it should retain the distinction between **audited monetary expenditure**, **physical social/environmental burden**, and **chosen shadow valuation**. Collapsing all three into one number would make controversial ethical choices look like accounting facts.

It must also avoid double counting. If a supplier's electricity price already contains a carbon tax and the same carbon damage is added again as a shadow cost, the ledger needs to show the tax as a transfer and the physical damage separately rather than blindly adding both as if they were independent harms.

Question seven — Traceable distribution

The evidence that transparency alone changes distributive decisions is weaker than the theoretical case for it.

Open-book accounting has been used in buyer-supplier relationships to expose cost structures and search jointly for savings. Kajüter and Kulmala's case research found successful uses but also failures arising from insufficient trust, weak accounting systems and asymmetric bargaining relations. [48] Cost visibility therefore does not eliminate power. A buyer who sees a supplier's cost may use that information to squeeze it rather than to make a collectively preferred distributional choice.

Participatory budgeting provides better evidence that making allocation explicit can matter. Gonçalves's study of Brazilian municipalities found adoption of participatory budgeting associated with changes in public spending and lower infant mortality, although adoption was not randomized and causal interpretation therefore depends on the study's identification strategy. [49]

Olken provides a randomized contrast. In 49 Indonesian villages, replacing representative meetings with plebiscites over local public goods substantially improved satisfaction, knowledge and perceived legitimacy, but changed the actual projects chosen much less. [50] That is a highly relevant result for the present proposal: **explicit choice can have a larger effect on legitimacy and information than on the distributive outcome itself**.

Cooperatives offer a simpler form of transparent distribution. USDA descriptions of patronage refunds show a practical “service at cost” mechanism: cooperatives cannot know exact annual costs in advance, so they charge prices during the year, calculate the eventual margin, retain what is needed for capitalization and return the balance according to patronage. [8] The rule does not eliminate distributive politics. It makes the rule inspectable.

Frameworks such as the Economy for the Common Good and doughnut economics extend accounting beyond financial profit toward social and ecological objectives. Evidence of implementation exists, but I found no primary causal evidence strong enough to conclude that adopting such a scorecard by itself systematically produces more equal or more sustainable outcomes. Research on municipal doughnut implementation is still largely about process, interpretation and institutional translation. [51]

Inference. Traceability is best understood as **constitutional infrastructure**, not as an outcome guarantee. It makes decisions contestable. It permits someone to ask: “Why did this user group pay this share? Why was this developer's bounty €10 million rather than €2 million? Why was this environmental value used?” That is a genuine gain even when the eventual political choice remains unequal.

Question eight — Cases at scale

Mondragon is useful but often misdescribed for this purpose. It is a large federation of worker cooperatives with democratic ownership, inter-cooperation and distinctive internal distribution rules, but it is not an economy of zero-profit firms selling everything at cost. Its businesses explicitly pursue competitiveness and profitability; its recent reports record substantial sales, profits, international activity and R&D. [52] Mondragon is therefore prior art for federated governance and distribution of surplus, not for eliminating surplus.

Agricultural cooperatives are closer. Patronage accounting allows them to operate on a service-at-cost principle while retaining working capital and investment funds. [8] US rural electric cooperatives likewise describe themselves as member-owned, not-for-profit providers of at-cost electric service, serving tens of millions of consumers through hundreds of local systems. [53] Their experience also shows why “at cost” cannot mean “charge only today's running expense”: networks require capital, reserves, maintenance and replacement.

Regulated utilities demonstrate cost-based pricing at still larger scale. FERC cost-of-service methodology explicitly includes operating expense, depreciation, taxes and a return on rate base in the revenue requirement. [54] The allowed return is not an accidental contamination of cost pricing. It finances capital and compensates investors. Modern regulation has consequently moved beyond simple cost reimbursement; Ofgem's RIIO framework combines revenue regulation with incentives, innovation and output obligations. [55]

This history is a warning for the proposal. Removing accounting profit does not remove the economic cost of capital. A cooperative can supply capital through retained patronage. A public system can obtain it through taxation or public borrowing. A private regulated firm receives an allowed return. Some institution always bears the opportunity cost and risk.

Shared digital infrastructure provides another relevant precedent. GÉANT links European national research and education networks through a membership and project-funded association. Its 2025 accounts show membership income alongside substantial project funding, and its governance has explicitly strengthened its cash-reserve policy to support resilience. [56] This looks much more like the proposed “interfaces as a commons” than a conventional retail platform, although it remains dependent on external public funding and member institutions.

Platform cooperatives show both promise and scale difficulties. Recent empirical work reports problems of member participation, capital raising and the strategic need for scale. A study based on interviews with leaders of platform worker cooperatives found both successes and recurring governance challenges; other feasibility work finds the model considerably easier in some professional or taxi markets than in capital- and logistics-intensive delivery markets. [57]

Planned-economy experience supplies the strongest caution against soft funding. Kornai's soft-budget-constraint analysis describes organizations whose expenditure ceases to be tightly constrained by earnings because they expect subsidies, soft credit or rescue. He links that expectation to weakened price responsiveness and inefficient behavior. [58] This is directly relevant to a central interface or bounty fund. A fund that routinely rescues overruns turns a reserve into an entitlement.

The comparison with Soviet-style planning should not be pushed too far. The proposed system is compatible with independent firms, entry, consumer choice and market-priced inputs. It is not material-

balance central planning. But the historical failure mode—administrators responding to shortages by enlarging budgets rather than forcing opportunity-cost choices—remains relevant.

Inference. Large-scale precedents support *pieces* of the system. The most successful ones share four characteristics: protected renewal funding, hard accounting boundaries, decentralized operating responsibility and some mechanism stronger than cost reimbursement for rewarding performance. The bad cases weaken one or more of those constraints.

Question nine — Composition and stability across chains

Spengler's 1950 analysis of vertical integration established the classic double-marginalization result: where successive monopolists each add a markup, the final price can exceed the vertically integrated monopoly price and output can fall further. [59] This strongly supports the intuition that independently added margins do not compose benignly.

But the formal statement "margins compound" requires a pricing convention. If each supplier applies an ad-valorem multiplier $1 + m_i$, then a chain does indeed multiply the factors:

$$P = C \prod_i (1 + m_i).$$

If each adds an absolute dollar margin, the effect is additive. The important economic result is not that every margin literally compounds exponentially; it is that successive pricing power can create cumulative wedges and double marginalization.

The bullwhip effect is a different instability. Lee, Padmanabhan and Whang showed how order batching, demand-signal processing, shortage gaming and price variation can make upstream orders more volatile than final demand. [60] Transparent at-cost prices do not automatically solve this because the amplification arises from inventory and information policies as well as price margins.

Coase's transaction-cost theory also refines the proposition about interface costs. Using the market requires finding prices, negotiating and enforcing transactions; organizations internalize transactions when market coordination becomes more costly than internal organization. [61] Williamson developed this into a broader theory involving asset specificity, uncertainty and contracting hazards. [62] These theories support the idea that costly interfaces discourage long chains. They do **not** imply that *any* positive interface cost mathematically imposes a finite maximum chain depth. A finite bound follows only when the value created by adding another layer is bounded relative to the additional cost.

Shared-cost allocation is another mature literature. Aumann and Shapley derive values for economies with continua of small participants. [63] Billera, Heath and Raanan applied the method to a real internal telephone network at Cornell. Their problem was almost exactly to allocate bulk and nonlinear service costs among many small users while satisfying cost recovery and defensible fairness properties; the resulting rates were actually implemented. [64]

This is important prior art for capacity allocation. A simple "capacity cost divided by the number of users a unit serves" is one defensible rule when users impose homogeneous requirements. Aumann–Shapley

allocation becomes attractive when loads are heterogeneous and cost is nonlinear. Neither method makes “fairness” objective: each embodies axioms about which incremental contribution should count.

Inference. The proposal composes cleanly in the *linear accounting layer*. Real-world stability requires another layer dealing with nonlinear capacity, inventory, uncertain demand, correlations among errors and transaction costs.

Question ten — Coexistence and transition

There is no theoretical reason an at-cost layer has to replace markets. The strongest version is a **public or cooperative option** embedded inside them. Conventional firms remain free to charge market prices. Suppliers that accept the at-cost protocol receive access to common infrastructure and become eligible for bounties. Consumers can compare the protocol's price with ordinary offers. That makes the protocol both a service and a reference price.

The precedents above already coexist with ordinary markets: cooperatives, cost-of-service utilities, research-network consortia, open-source businesses and public procurement all do so. [65] The HIF was explicitly designed as an optional alternative alongside ordinary pharmaceutical rights rather than universal abolition of patents. [66]

EU law imposes important boundaries. Article 107 TFEU treats selective advantages financed through state resources that distort or threaten competition and affect intra-EU trade as state aid unless an exemption or compatibility rule applies. [67] Under the *Altmark* criteria, compensation for a clearly defined public-service obligation can fall outside State-aid classification when the compensation parameters are objective and set in advance, compensation does not exceed the necessary cost plus reasonable profit, and provider selection or efficient-firm benchmarking satisfies the fourth condition. [68]

The EU's current SGEI framework was updated by Commission Decision 2025/2630, while Regulation 2023/2832 provides a de-minimis regime for certain SGEI compensation, with a €750,000 ceiling per undertaking over three years under that instrument. [69] A large bounty scheme would often fall outside de minimis and would need to be structured through procurement, R&D/state-aid rules, SGEI rules or another lawful basis.

There is a second, less obvious competition-law issue: **open-book pricing among competitors can itself be dangerous**. The Commission's horizontal-cooperation guidelines warn that exchanges of commercially sensitive current or future price and quantity information can facilitate coordination. They suggest arrangements such as independent trustees, clean teams and aggregation where data sharing has a legitimate purpose. [70]

So “transparent” should not mean “every rival sees every supplier's real-time detailed cost sheet.” A public system can make the *derivation and auditing rule* transparent while letting an independent trustee hold commercially sensitive source data. Historical and aggregated benchmark data can be public.

Inference. Transition is feasible, and coexistence is likely safer than wholesale replacement. A public-option structure also creates a powerful empirical test: if ordinary firms consistently beat the at-cost provider on total cost and quality without requiring monopoly rents, they remain. If the at-cost option substantially widens access, that becomes observable rather than ideological.

Prior art and novelty

Prior art	What it already provides	Closeness to the proposal	What remains different
Arrow, 1962 [71]	Efficient near-free use of information; separation of innovation reward from user charge	**Very high conceptually**	No general operational pricing/funding protocol
Leontief, 1936 [19]	Recursive inter-industry cost/input system	**Very high for stacking**	Descriptive/analytic IO system, not governance or bounty design
Hawkins–Simon [20]	Conditions for productive non-negative IO systems	**High mathematically**	Does not solve incentives or dynamic discovery
Sraffa, 1960 [21]	Production prices and explicit distributive variable	**High as critique**	Does not advocate zero-profit bounty architecture
Hotelling, 1938 [38]	Marginal-cost pricing plus external fixed-cost finance	**High**	Fixed investment and governance remain unresolved
Coase, 1946 [39]	Critique of subsidized marginal-cost pricing; investment/value revelation	**High as objection**	No innovation-bounty layer
Oi, 1971 [40]	Entry fee plus marginal use price	**High for tariff form**	Entry fee can itself restrict distribution
Ramsey / public-utility pricing [41]	Second-best fixed-cost recovery	**Medium**	Intentionally retains markups
Kremer patent buyouts [72]	Separates innovation reward from monopoly exploitation	**Very high for bounty**	Patent-specific; auction/collusion problems
Shavell–van Ypersele rewards [24]	Formal prizes versus IP	**Very high**	Government still must estimate reward
Health Impact Fund [27]	Near-cost medicine price plus outcome-related reward	**Extremely high in one sector**	Proposed rather than demonstrated at full scale
Pneumococcal AMC [12]	Advance reward separated partly from long-run access price	**High**	Reward is contracted procurement rather than general bounty

Prior art	What it already provides	Closeness to the proposal	What remains different
Agricultural patronage coops [8]	Service-at-cost accounting with ex-post surplus true-up	**Very high operationally**	Rewards not separated as innovation bounties
Utility cost-of-service regulation [54]	Audited cost stack, depreciation, capital recovery	**High**	Known cost-plus incentive problems; retains return
Aumann–Shapley / Billera et al. [7]	Fair allocation of nonlinear common costs	**Very high for capacity/common costs**	No innovation reward
Ostrom [43]	Nested commons governance	**High for institutional layer**	Not a pricing system
Benkler / open source [73]	Non-proprietary information production with non-price incentives	**High for low-MC goods**	Financing remains heterogeneous
Environmental IO [74]	Recursive embodied environmental accounts	**High for external costs**	Valuation remains normative
Participatory budgeting [75]	Explicit collective allocation decisions	**Medium for distributional governance**	Not integrated with production prices
GÉANT-type infrastructure federation [76]	Shared digital infrastructure, membership and public funding, reserves	**High institutional analogy**	Not universal at-cost chain protocol

Assessment of novelty. The proposal is not a new theory that profit margins create exclusion, not a new theory of recursive cost pricing, and not a new proposal to pay innovators separately. Arrow alone is enough to reject a claim of novelty for the basic reward/access separation. [71]

The potentially original object is the **protocol-level synthesis**:

$$\boxed{\text{audited recursive cost price}} + \boxed{\text{causal capacity allocation}} + \boxed{\text{commons-funded interfaces}} + \boxed{\text{separate value rewar}}$$

I found no source in this review that specifies that complete combination as a general interoperable system. The strongest established relatives are Arrow + Leontief + cooperative patronage accounting + Aumann–Shapley common-cost allocation + prize/HIF mechanisms.

The seven formal findings tested

Formal finding	Verdict	Literature test and necessary qualification
Capacity pricing recovers step costs to within one capacity unit.	**Supported under stated premises; refine economically.**	If capacity is bought in indivisible blocks and the tariff allocates the cost of the required integer number of blocks, the one-block bound follows arithmetically. Peak-load, club and nonlinear-cost literature supports separating capacity from marginal use. [77] But "cost per block divided by users served" works cleanly only for sufficiently homogeneous demand. With heterogeneous peak loads, coincidence factors and reliability requirements, causal capacity shares can differ greatly.
Volume-based allocation of fixed costs causes a death spiral.	**Refined, not universally true.**	A feedback mechanism is real: when fixed network cost is recovered mainly through volumetric charges, users who reduce or bypass volume can leave a larger fixed requirement to remaining sales, raising volumetric rates and encouraging more bypass. Utility rate-design literature treats this as a genuine risk. [78] But a spiral requires sufficiently elastic demand/substitution and particular regulatory responses. Volume recovery of fixed cost does not mathematically imply collapse.
At-cost stacking conserves cost exactly.	**Supported algebraically; refine accounting boundary.**	In a linear chain where every upstream payment is counted exactly once and no margin is introduced, recursive stacking preserves the sum of primary costs. This is the core Leontief identity. [19] In practice, joint production, depreciation, taxes, imports, subsidies, shared overhead and environmental accounts require explicit conventions; otherwise "conservation" can double-count or omit components.
Margins compound across layers.	**Supported for multiplicative markups; broader result supported.**	Ad-valorem margins multiply mechanically. Spengler's double-marginalization result establishes the economically important version: sequential market power can produce cumulative price wedges and lower output. [59] Absolute markups add rather than multiply, so the formal theorem should state the markup convention.
Any positive cost per interface limits chain depth.	**Contradicted as a universal theorem.**	Coase and Williamson support the weaker proposition that transaction/interface costs influence organizational boundaries and make extra layers less attractive. [79] A strictly positive per-link cost creates a finite maximum depth only if the benefit available from additional depth is itself bounded or eventually falls below that cost. Without that premise, no finite-depth theorem follows.

Formal finding	Verdict	Literature test and necessary qualification
Step errors add across layers unless reserves are pooled.	**Refined.**	Worst-case deterministic errors can add. Independent mean-zero forecast errors need not: positive and negative errors partly cancel, and statistical pooling can reduce reserve needs when correlations are below one. Pooling therefore improves risk sharing, but it is not necessary for all cancellation and can create moral hazard if layers expect automatic bailout. This result should be rewritten probabilistically, with explicit assumptions about correlation and loss allocation.
Covering running costs without renewal runs a service down.	**Supported.**	Cost-of-service regulation explicitly recognizes depreciation and capital requirements in addition to current operations. [54] A tariff that covers labor, electricity and cloud bills but not replacement of servers, software refactoring, security remediation or other capital renewal is not genuinely at cost over the asset's life. The appropriate concept is *long-run sustainable cost*, not cash operating expense.

Two formal conclusions deserve special emphasis.

First, **the fifth finding should be changed**. The literature supports “interface costs discourage and can bound chain depth,” not “any positive interface cost necessarily bounds it.” The missing premise is bounded incremental value.

Second, **the second finding should also be weakened**. A death spiral is a dynamic possibility, not an accounting identity. The theorem needs assumptions about demand elasticity, alternatives, fixed-cost recovery rules and rate adjustment.

The strongest of the seven are the third, fourth and seventh. They correspond directly to mature literatures in input-output accounting, vertical pricing and infrastructure finance.

Sizing and governing the bounty

The bounty is the most important unsolved component because it must replace several functions currently bundled into profit.

What the bounty should measure

The simplest proposal,

$$B \propto Q$$

where Q is at-cost demand, is inadequate. Quantity at a low price measures use, not surplus. The more useful target is incremental social value:

$$V_j = \sum_i (\text{outcome}_{ij} - \text{counterfactual outcome}_{ij}) \times \text{social value per outcome} - \text{incremental social costs}.$$

The bounty might then be constrained by

$$B_j = \min\{B_{\max}, \alpha \widehat{V}_j\}.$$

That resembles the logic of the HIF, where rewards are proportional to independently estimated incremental health impact, and of prize theory more generally. [80] It should not be read as a universal formula. The counterfactual and social valuation are often the hardest quantities in the entire system.

For commercial-type goods, a second route is to construct a **counterfactual demand curve** even though the actual selling price remains at cost. Users can make scarce, binding commitments indicating what they would sacrifice to obtain the solution. Kremer's auction idea shows why actual economic stakes are preferable to surveys: cheap statements invite exaggeration. [72]

A third route is **advance problem contracting**. A sponsor states: “We will pay €X if someone achieves Y by date Z and makes the solution available under these access conditions.” This is the AMC/prize model. Its advantage is certainty for innovators; its weakness is that the sponsor must specify the right target before seeing the solution. [81]

A fourth is **decentralized fund allocation**, including matched contributions. Quadratic funding is attractive where many users each have partial information about which public goods matter, but it requires anti-collusion and identity safeguards and does not itself solve the problem of poor users having less money to signal value. [31]

A practical hybrid

For a serious pilot, I would not make the bounty a single demand multiplier. A more defensible structure is:

$$B = B_{\text{milestone}} + B_{\text{adoption}} + B_{\text{impact}} + B_{\text{efficiency}},$$

subject to a pre-announced cap.

The **milestone** component pays for objectively verified creation and reduces financing risk. The **adoption** component captures decentralized information about whether people actually use the solution. The **impact** component prevents trivial high-volume use from dominating socially important low-volume solutions. The **efficiency** component rewards achieving those outcomes at lower lifetime resource cost, countering the cost-plus incentive problem identified by Laffont and Tirole. [37]

No one component should be controlled by the provider.

Who should pay

The payer should follow the benefit boundary rather than the transaction boundary.

A solution benefiting one organization can be paid for by that organization. A solution benefiting a sector can be funded through a sector levy or purchasing consortium. A non-excludable national public good can

be tax-financed. A global public good, such as knowledge relevant to infectious disease, requires multinational or philanthropic finance because one country's demand systematically understates spillovers.

This is not an accounting fact. It is a distributive choice.

A fixed annual fund has a major advantage: it creates a **hard budget constraint**. HIF-style proportional sharing also makes reward claims compete with each other rather than turning every positive impact estimate into an unlimited liability. [27] But a fixed pool makes the reward per unit of impact endogenous; innovators face uncertainty about how many competitors will qualify.

Governance

The bounty authority should be institutionally separate from both the cost auditor and the providers. At minimum it requires an independently appointed board, published eligibility rules, ex-ante reward formulas where possible, conflict-of-interest declarations, independent evaluation, auditable raw data, random audits, anti-collusion rules, appeal procedures, clawbacks for fraud or material measurement error, and explicit rules for intellectual-property release.

The mechanism should also distinguish **failure** from **fraud**. Innovation requires unsuccessful experiments. A bounty regime that punishes honest technical failure will cause excessive conservatism.

The cost auditor should verify the price but should **not simply reimburse whatever cost is incurred**. Benchmarking, competitive purchasing, pre-agreed cost categories and efficiency targets are needed to prevent gold-plating. Averch–Johnson and Laffont–Tirole are the relevant warnings. [10]

Bounty governance should have a hard no-bailout rule or a formally specified emergency procedure. Kornai's soft-budget analysis predicts what happens when participants learn that overruns will routinely be socialized. [58]

Finally, the system should not reveal competitively sensitive current cost information indiscriminately. In the EU, an independent audit body or trustee could publish certified aggregate cost components while keeping provider-specific underlying data segregated. That architecture is compatible with the Commission's guidance on legitimate data pools and clean-team arrangements. [82]

Minimal pilot design

The cleanest first test is **open-source software for public institutions**. Software combines high initial problem-solving cost, very low reproduction cost, measurable usage, relatively rapid iteration and the possibility of giving every participant the same code. It avoids many of the clinical, manufacturing and patent complications of a medicine pilot while testing almost every economic mechanism in the proposal. Arrow's information argument applies directly. [71]

A suitable concrete pilot would be a **shared public-health analytics platform** used voluntarily by municipalities, public-health institutes, universities or healthcare agencies. The pilot should solve real analytic problems—data validation, surveillance workflows, reproducible dashboards, statistical pipelines or privacy-preserving data interfaces—rather than create another generic administrative platform.

Institutional structure. A consortium of perhaps a few dozen participating organizations establishes a legally separate commons operator. The numbers are design suggestions, not empirical optima. The operator owns or stewards identity, standards, package registry, security requirements, interoperability tests, telemetry and common documentation. Individual software suppliers remain independent and compete to solve published problems.

Price. Every certified component has a visible long-run sustainable cost schedule:

$$P = F_{\text{capacity}} + r_c C + \sum_k q_k c_k.$$

Here $q_k c_k$ is auditable metered use such as cloud compute, external API calls or storage. $r_c C$ is the user's causal share of reserved support, operations or capacity. The common operator finances basic interoperability, identity, shared security standards and the bounty machinery from the consortium fund rather than charging a toll every time one component calls another.

Development expenditure already rewarded by a bounty is **not** added again to per-copy prices. Maintenance that is genuinely incremental and recurring is included. Major renewal is funded through a separate depreciation/renewal account. This prevents both double payment and the seventh formal failure.

Cost recursion. Suppliers report purchased inputs to the audit trustee. The public ledger shows certified categories rather than competitors' raw invoices:

$$p_j = c_j + \sum_i a_{ij} p_i.$$

Every component therefore exposes its recursively embodied cost. The pilot can compare the resulting bottom-up values with market SaaS prices and with coarse economy-wide IO benchmarks. [83]

Problem discovery. Participants receive a limited annual allocation of *problem tokens*. Tokens cannot be converted to cash and must be allocated among proposed problems. Because the budget is scarce, supporting one problem has an opportunity cost. This produces a stronger demand signal than clicking “interested.”

Organizations proposing a problem must additionally commit to installing or testing a qualifying solution if it meets a published specification. That makes expressions of demand partly binding.

Bounty competition. For each sufficiently supported problem, the consortium publishes the specification, evaluation method and maximum available bounty. Multiple teams may compete. The bounty has four tranches: a modest verified milestone payment; payment for independent security/quality acceptance; a usage/adoption payment after real deployment; and an impact/efficiency payment after an evaluation period.

A winning supplier must license the funded core so that all consortium members can use, modify and redistribute it under the pilot's agreed terms. Suppliers remain free to sell integration, customization, hosting and unrelated proprietary products outside the common layer.

Capacity allocation. Homogeneous hosting resources can use the simple capacity-unit rule in the proposal. Where users generate strongly heterogeneous peak loads, the pilot should simultaneously compute an Aumann–Shapley-type allocation as a comparison. Billera and colleagues provide a real precedent for applying that framework to nonlinear common service costs. [7]

Reserve. Annual tariffs are set ex ante. Forecast errors inside a predefined band are absorbed by a pooled reserve; deviations outside the band trigger an explicit re-rating procedure. The reserve has a maximum size and cannot automatically rescue supplier overruns. The consortium's reserve policy should distinguish aggregate random forecast error from provider-specific inefficiency.

Experimental evaluation. The pilot should be designed to answer questions rather than merely demonstrate feasibility. Where operationally possible, stagger rollout between otherwise comparable organizations. Track:

Outcome	What it tests
Cost per active user	Whether recursive at-cost distribution lowers access cost
Number and diversity of adopting organizations	Maximum-distribution claim
Incremental use after price reduction	Size of the access effect
Supplier entry and number of proposals per problem	Whether bounties preserve innovation competition
Time from problem publication to usable solution	Discovery efficiency
Defects, uptime, security incidents	Whether quality deteriorates
Lifetime maintenance and renewal cost	Whether low headline prices conceal future cost
Cost forecast errors and reserve draw	Stability of annual tariffs
Administrative/audit cost	Whether the mechanism itself becomes an expensive interface
Distribution of bounty recipients	Concentration/capture
User willingness to commit scarce problem tokens	Revealed demand
Counterfactual outcome improvement	Social value rather than mere adoption
Conventional SaaS/procurement comparator	Opportunity cost of using the new system

The most revealing failure criterion is not “the at-cost system costs more than its own prediction.” It is: **once common administration, auditing, bounty expenditure, renewal and risk are all counted, does it deliver more useful access per euro than conventional procurement?**

That prevents costs from disappearing merely because they have moved to a central ledger.

Risks, open questions, and final verdict

The first risk is **confusing cost with value**. Recursive prices answer “what resources did this require?” They do not answer “was it worth doing?” Hayek's point survives intact. [84] The system therefore needs a value-revelation mechanism alongside the cost system.

The second is **confusing demand with willingness to pay**. Maximum distribution intentionally destroys some of the price variation that ordinarily reveals willingness to sacrifice other resources. Demand at €0.01 contains less information than a real choice between €100 and an alternative use of €100. The bounty architecture must deliberately reconstruct that information through commitments, auctions, outcome measurement or competitive budget allocation. Kremer and the HIF are different attempts to solve exactly that problem. [85]

The third is **cost inflation**. A provider with guaranteed pass-through has weak incentives to economize. This is the fundamental cost-plus problem. Competitive tendering, benchmark costs, fixed-price components and output rewards are not optional add-ons; they are core parts of a viable design. [10]

The fourth is **capital and uncertainty**. “No profit” cannot mean “no return to anyone for bearing risk.” Someone must finance work before the bounty is known, absorb failed projects, maintain liquidity and fund assets between expenditure and reimbursement. A public fund can bear that risk, as can cooperative capital or private investors paid through an explicit risk allowance. Hiding the cost is not the same as eliminating it. Utility accounting's inclusion of depreciation and capital return illustrates the point. [54]

The fifth is **capture of the bounty authority**. Once innovation rents are transferred from monopoly prices to a public or collective fund, lobbying incentives move with them. A €1 billion monopoly rent attracts efforts to win the monopoly; a €1 billion prize pool attracts efforts to influence the prize criteria. Kremer explicitly recognized collusion as a problem in patent buyouts. [72] Transparency helps detect capture but cannot eliminate it.

The sixth is **soft budgets**. Central funding of interfaces is efficient only if interface designers face an actual limit. Otherwise each provider can declare another shared feature “common infrastructure.” Kornai's analysis predicts weakened discipline when actors expect excess cost to be absorbed elsewhere. [58] Commons funding needs a budget, prioritization procedure and credible refusal mechanism.

The seventh is **underinvestment in renewal**. This is especially dangerous politically because keeping today's price low by postponing maintenance is immediately attractive while the failure appears later. Renewal should therefore be a protected first-class cost rather than an optional surplus allocation. [54]

The eighth is **allocation under scarcity**. At-cost prices can generate excess demand whenever short-run capacity is fixed. The proposal needs explicit rules for queues, priorities, reservations, auctions or temporary scarcity prices. Otherwise shortage becomes the unpriced rationing mechanism.

The ninth is **measurement capture and Goodhart effects**. A bounty based on health outcomes induces optimization toward measured health outcomes; one based on users induces acquisition of users; one

based on reported savings induces accounting changes. The HIF authors themselves recognize that reward assessment must be designed so that the easiest way to obtain payments is genuinely to improve health. [27] No general mechanism makes this problem disappear.

The tenth is **legal exposure**. Public subsidy, public procurement, IP licensing and competitive information sharing all sit inside existing legal systems. In the EU, both State-aid rules and Article 101 concerns about competitively sensitive data matter. [86] This favors an opt-in pilot operated through conventional procurement or a clearly defined public-service structure, with independent cost auditing.

There are also several genuinely open research questions. Can counterfactual willingness to pay be elicited without putting exclusion back into the final price? Can a bounty formula reward breakthrough innovation while remaining predictable enough to finance risky R&D? How much administrative cost does recursive auditing add per layer? Does cost transparency facilitate entry, or expose small suppliers to stronger buyers? How should indivisible capital be allocated when different users cause capacity jointly? Which externalities can be legitimately monetized and which should remain physical constraints? How large must reserves be when errors across layers are correlated? Can exit discipline be retained when a common service becomes systemically indispensable?

The most consequential conceptual revision I would make to the proposal is therefore this:

> Do not define profit as the bounty.

Instead, define **innovation rent as one function currently performed by profit**, and ask explicitly where every other function of profit goes.

A complete decomposition would be:

$$\text{ordinary market margin} \longrightarrow \begin{cases} \text{innovation reward} & \to \text{bounty}, \\ \text{risk bearing} & \to \text{explicit risk finance}, \\ \text{capital renewal} & \to \text{depreciation/renewal account}, \\ \text{scarcity signal} & \to \text{capacity allocation mechanism}, \\ \text{cost-efficiency incentive} & \to \text{performance contract}, \\ \text{discovery/entry} & \to \text{open competition}, \\ \text{residual surplus} & \to \text{explicit distributive rule}. \end{cases}$$

That decomposition is, in my assessment, the strongest version of the idea.

Final verdict. The literature does not support an economy-wide rule saying simply “set every price equal to accounting cost and replace all profit with a demand bounty.” It does support something more precise and potentially powerful: **for goods whose marginal use cost is well below their value, especially information goods, separate the price of access from the reward for creation wherever the reward, capital and governance mechanisms can be specified credibly.** Arrow provides the core welfare logic. Leontief provides the cost recursion. Cooperatives and utility regulation provide cost-recovery institutions. Prize theory, patent buyouts, AMCs and the HIF provide reward mechanisms. Ostrom provides governance principles. The incentive-regulation and planning literatures explain why these pieces cannot simply be assembled without hard budget constraints, competition, scarcity signals and decentralized discovery. [87]

The proposal's strongest claim is consequently not that markets are unnecessary. It is that **the market currently bundles several logically distinct functions into one markup, and for low-marginal-cost goods there may be large gains from unbundling them**. That claim is strongly grounded in prior theory and worth testing empirically.

Source list

The list below emphasizes original papers, books, official institutional evaluations and current legal texts rather than secondary summaries.

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